

Work Instruction — BV-12 Assembly

Manufacturing Engineering · 2025-06 · Revision D

1. Scope

This instruction covers the assembly of the BV-12 valve assembly from received sub-components to the point of final inspection. It applies at all sites holding the assembly in scope of work.

Field	Entry
Instruction	WI-BV12
Revision	Revision D
Issued	2025-06
Supersedes	Revision C
Applies to	BV-12 valve assembly

This revision is issued under CR-91. The revision introduces a defined tightening sequence at the housing interface and a recheck requirement. No other provision of the instruction is altered by this revision.

Operators shall work to the current revision only. Superseded revisions are withdrawn from the line and retained by Manufacturing Engineering for record purposes. Where a superseded revision is found at the workstation it shall be returned to Manufacturing Engineering and replaced.

2. Tooling

Tooling required for the assembly is listed below. Tooling shall be within calibration and shall carry a current calibration label. Tooling outside calibration shall not be used and shall be withdrawn to the calibration cabinet.

Item	Specification	Calibration
Torque wrench	Range covering the specified value	6 months
Assembly fixture	Fixture FX-BV12-02	Annual verification
Seal insertion tool	Tool ST-14	Visual check each shift
Depth gauge	Per drawing tolerance	6 months

The torque wrench shall be selected such that the specified value falls within the middle of the instrument range. A wrench used near the limit of its range shall not be used for this assembly.

Calibration status shall be confirmed at the start of each shift and recorded on the shift log. Where a tool is found outside calibration during a shift, assemblies built since the previous confirmed check shall be quarantined pending assessment.

Fixtures shall be cleaned at the end of each shift and inspected for wear at the interval stated in the fixture maintenance schedule. Fixture wear affects seating and shall be reported to Manufacturing Engineering rather than corrected at the workstation.

3. Sub-assembly

The sub-assembly stage prepares the valve body and the internal components for housing assembly. Components shall be drawn from released stock only and shall carry the batch identifier on the accompanying documentation.

Clean all mating faces before assembly. Faces shall be free of machining residue, handling marks and protective compound. Use only the cleaning agent specified in the materials list; substitution is not permitted.

Step	Operation	Check
3.1	Clean mating faces	Visual
3.2	Inspect seal seating groove	Visual and gauge
3.3	Fit seal using insertion tool ST-14	Visual
3.4	Verify seal seating depth	Depth gauge
3.5	Fit internal components per drawing	Visual

The seal shall be fitted using the insertion tool. Fitting by hand or with any other tool is not permitted, as the seal may be stretched beyond its recovery limit without visible indication.

Verify seating depth after fitting. A seal seated outside the drawing tolerance shall be removed and discarded; a removed seal shall not be refitted. Record the check on the build record.

Where the seating groove shows damage, the valve body shall be withdrawn and reported. Damage to the groove cannot be corrected at the workstation.

Sub-assembly is complete when all checks in the table above are recorded against the build record. The assembly shall not proceed to housing assembly with an outstanding check.

4. Housing assembly

The housing is secured to the valve body with the M8 housing bolts supplied in the assembly kit. Confirm before starting that the kit is complete and that the bolts carry the correct finish; bolts of the correct dimension but incorrect finish are supplied against other assemblies in the series and are not interchangeable.

Present the housing to the valve body squarely, taking care not to displace the seal fitted at the previous stage. Seat the housing by hand until the flange faces meet. A gap at the flange face indicates that the seal has been displaced or that debris is present at the interface; in either case the housing shall be removed and the interface reinspected before proceeding.

Engage all fasteners by hand before applying any torque. Hand engagement confirms thread alignment and prevents cross-threading, which cannot reliably be detected once torque has been applied. A fastener that does not engage freely by hand shall not be forced; withdraw it, inspect the thread, and replace the fastener if the thread shows damage.

Fasteners shall be fitted dry. Do not apply lubricant, sealant or thread locking compound at this interface unless expressly directed by a concession raised against the specific build. Lubricant alters the relationship between applied torque and achieved clamp load, and a lubricated fastener tightened to the specified value will be over-tightened at the joint.

Tighten in a diagonal sequence. Set it to 40 Nm. Recheck after 30 minutes.

The recheck is required because the seal relaxes after initial compression and the clamp load falls. The recheck shall be recorded on the build record against the assembly serial. An assembly released without a recorded recheck is not conforming and shall be returned for the recheck to be performed.

Step	Operation	Record
4.1	Confirm kit complete and finish correct	Build record
4.2	Seat housing, confirm flange contact	Visual
4.3	Engage all fasteners by hand	Visual
4.4	Apply specified torque in sequence	Build record
4.5	Recheck at the stated interval	Build record

Where the recheck finds a fastener below the specified value, the sequence shall be reapplied in full and a further recheck performed at the stated interval. A fastener requiring a third application shall be withdrawn and the assembly reported to Manufacturing Engineering.

5. Inspection

Final inspection confirms that the assembly conforms to the drawing and that all checks required by this instruction have been recorded.

Check	Method	Acceptance
Flange contact	Visual and feeler	No measurable gap
Fastener torque record	Build record	All entries present
Recheck record	Build record	Present, at stated interval
Seal seating	Build record	Within drawing tolerance
External condition	Visual	Free of damage and residue

An assembly failing any check shall be quarantined and reported. Rework shall be performed only under a rework instruction raised for the specific condition; rework at the workstation is not permitted.

Conforming assemblies shall be labelled with the assembly serial and the build record reference and passed to packing. The build record shall be retained for the period stated in the records procedure.